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EDITORIAL



AI literacy and competency: definitions, frameworks, development and future research directions

1. Introduction

The rapid integration of artificial intelligence (AI) into daily life – from generative tools like ChatGPT to predictive algorithms shaping our newsfeeds – has made AI literacy and competency urgent educational priorities (Chiu, 2024, 2025). This is reflected in all the frameworks proposed by various organizations such as the Organization for Economic Co-operation and Development (OECD)/European commissions and the United Nations Educational, Scientific and Cultural Organization (UNESCO, 2025). While often used interchangeably, AI literacy and AI competency are related but distinct concepts, operating at different levels of understanding and skill.

2. AI literacy vs AI competency

2.1. AI literacy: the foundation of awareness

AI literacy refers to a foundational conceptual understanding of AI. It focuses on knowledge, critical thinking, and ethical awareness rather than technical skill (Chiu et al., 2024; Zhou et al., 2025). AI-literate individuals can explain what AI systems are (e.g., machine learning, neural networks), identify their real-world applications (like recommendation engines or facial recognition), and discuss societal implications such as bias, privacy risks, or economic disruption. They may be able to design, manage, engage, and create with AI (OECD, 2025). The OECD framework structures AI literacy around four interaction domains: (i) Engaging with AI: Recognizing AI use and critically evaluating outputs (accuracy, bias, relevance); (ii) Creating with AI: Collaborating with AI for ideation, content generation, and problem-solving while addressing ownership/attribution; (iii) Managing AI: Strategically delegating tasks to AI to enhance human work (e.g., automating logistics to focus on team building); and (iv) Designing AI: Understanding the principles behind AI systems (data, design choices) to explore fairness and impact, even without building AI products.

This literacy empowers individuals to interact with AI responsibly – questioning outputs, recognizing limitations, and making informed decisions as consumers, citizens, or professionals (Chiu, 2025). Crucially, AI literacy is for everyone: students, teachers, policymakers, and the public. It does not require sophisticated coding ability but demands curiosity and critical engagement. AI literacy focuses on knowledge, understanding, and application of AI.

2.2. AI competency: the realm of action

AI competency, in contrast, denotes practical proficiency in using, engaging, interacting, developing, or managing AI systems in real-world contexts (Chiu et al., 2024, 2025; Zhou et al., 2025). It moves beyond knowledge and understanding into capacity (Chiu et al., 2024). AI-competent individuals should have the confidence and ability to use AI to complete specific tasks ethically, healthily, responsibly, and productively (Zhou et al., 2025). For example, AI-competent students could effectively use AI to support their self-regulated learning, and AI-competent programmers could effectively integrate AI into coding tasks to make computer applications. In other words, they are confident and have great capacity to clearly explain how AI technologies work and impact

society. They are able to use them in an ethical and responsible manner and to effectively communicate and collaborate with them in a specific setting (Chiu et al., 2024, 2025). They should have self-reflective mindsets to critically evaluate their AI tools and knowledge. AI competency is contextual and focuses on how well individuals use AI in beneficial ways.

Overall, AI literacy is about knowledge and understanding; AI competency is about skillful application and optimization. While literacy asks, “What does this AI do?” competency asks, “How can I make this AI work better?” AI competency fundamentally depends on AI literacy; you cannot effectively optimize or troubleshoot an AI system (“How?”) without first understanding its core functions and constraints (“What?”). Literacy is the compass; competency is the engine. Thus, while literacy provides the necessary knowledge, competency represents the higher-level skill of applying that knowledge to achieve superior outcomes and make the AI work ethically, effectively, and healthily for you.

3. The interdisciplinary: how diverse literacies form the foundation of AI literacy

AI is not a monolithic technology emerging from a single discipline. It is a profoundly interdisciplinary field, i.e., the foundational understanding of what AI is, what it does, and its implications cannot exist in isolation. It is intrinsically woven from a rich tapestry of other essential literacies (Chiu, 2025). Understanding AI requires navigating this complex intersection, where competence in one area informs and depends on understanding in others. We discuss ten literacies relating to and underpinning AI literacy:

1. **Mathematical Literacy:** AI, particularly machine learning (ML), relies heavily on mathematical frameworks. Linear algebra underpins data representation and transformations in neural networks. Calculus is crucial for optimization algorithms that train models (like gradient descent). Probability theory is fundamental for reasoning under uncertainty, Bayesian inference, and evaluating model confidence. Without mathematical literacy, core ML concepts remain abstract and impenetrable.
2. **Data Literacy:** AI systems are fundamentally data-driven. Data literacy – encompassing understanding data types, sources, collection methods, cleaning, manipulation, visualization, and fundamental analysis – is paramount. It enables individuals to critically assess the quality of the data feeding AI, understand biases inherent in datasets, interpret the results AI produces, and recognize the limitations of data-driven conclusions.
3. **Ethical Literacy:** AI raises profound ethical questions about bias, fairness, privacy, accountability, transparency (explainability), autonomy, and societal impact. Ethical literacy provides the philosophical frameworks (consequentialism, deontology, virtue, and ethics) and critical reasoning skills necessary to identify these dilemmas, analyze potential harms and benefits, and advocate for responsible AI development and deployment. It could be the most critical companion literacy.
4. **Media Literacy:** AI is surrounded by hype, misinformation, and powerful narratives. Media literacy empowers individuals to analyze claims about AI capabilities critically, identify biases in outputs or reporting, discern deepfakes or AI-generated text, understand the motivations of different stakeholders (companies, governments), and evaluate the broader societal and political implications of AI adoption. Media literacy focuses on understanding how AI shapes and is represented in information channels.
5. **Computational Literacy:** Understanding fundamental computing concepts (algorithms, logic, hardware/software interaction, complexity) is essential to grasp how AI systems process information and execute tasks. It demystifies the “black box” at a basic level, explaining sequential processing and decision-making structures inherent in AI models.
6. **Linguistic Literacy:** Natural language processing is a cornerstone of modern AI (chatbots, translation, sentiment analysis). Linguistic literacy provides the understanding of syntax, semantics,

pragmatics, morphology, and phonology that allows AI to parse, generate, and manipulate human language. It's crucial for understanding how large language models work, the nuances of prompt engineering, and the challenges AI faces with ambiguity, sarcasm, or cultural context.

7. **Visual Literacy:** With the rise of computer vision and generative AI models, understanding visual information – composition, color theory, symbolism, spatial relationships, and abstraction – is increasingly important. Visual literacy helps individuals interpret AI-generated imagery, understand how AI sees and analyzes visual data, and critically evaluate the realism or potential manipulations in synthetic media.
8. **Domain-Specific Literacy:** AI is applied in a context. Effective AI literacy within a specific field (e.g., education, healthcare, law, finance, engineering, or art) requires deep understanding of that domain's knowledge, practices, terminology, and existing challenges. A medical AI tool requires medical literacy to understand its purpose, interpret its outputs correctly, and identify potential clinical risks or biases. AI literacy is integrated with domain expertise to enable meaningful application.
9. **Scientific literacy:** the ability to understand, evaluate, and apply scientific knowledge directly supports AI literacy by enabling critical engagement with AI concepts like machine learning, data ethics, and algorithmic bias. Both require evidence-based reasoning, skepticism of unverified claims, and ethical reflection. A strong foundation in scientific thinking empowers individuals to navigate AI advancements responsibly and participate in shaping their societal impact.
10. **Design literacy:** the ability to understand and apply design principles, aligns closely with AI literacy. Both involve critical thinking, problem-solving, and evaluating choices—whether in design or AI systems. Design thinking's iterative, user-centered approach mirrors AI literacy's focus on ethical, effective AI use and development. Recognizing design principles in AI interfaces (e.g., usability, bias mitigation) enhances AI literacy. Conversely, AI literacy expands design literacy by incorporating data-driven decision-making. Together, they foster holistic, human-centered innovation across disciplines.

These literacies do not exist in silos; they constantly interact and inform each other within the sphere of AI literacy. For example,

- Understanding algorithmic bias (computational/ethical/design) requires analyzing training data (data) for skewed distributions (mathematical) that might reflect societal prejudices (media/ethical/design).
- Prompt engineering (linguistic) for an image generator (visual) relies on understanding how the model encodes concepts (computational/mathematical/scientific) and is constrained by its training data (data).
- Evaluating a medical diagnostic AI (domain-specific) necessitates understanding its reliability (mathematical), potential data limitations (data), and ethical implications of false positives/negatives (ethical).

AI literacy is not merely adding a new technical skill; it is about developing an integrated perspective (Chiu, 2025). It requires weaving together threads of understanding from computation, mathematics, statistics, data handling, language, visual perception, ethics, critical thinking, human cognition, and specific application domains. Gaps in any of these foundational literacies create vulnerabilities – misunderstanding capabilities, misinterpreting outputs, overlooking biases, or failing to grasp societal consequences. By recognizing and cultivating this interdisciplinary nature of AI literacy, individuals can build the robust understanding necessary to navigate the AI-driven world thoughtfully, critically, and responsibly, forming the essential foundation upon which AI competency can then be built.

4. The development of AI competency

Nurturing AI competency begins with establishing a foundational understanding of AI literacy – grasping core concepts, capabilities, limitations, and ethical implications (Chiu, 2025; Zhou et al., 2025). Once this base is solidified, progression requires deliberate practice and contextual application. Identify specific, ethical opportunities within your domain (e.g., research analysis for researchers, report drafting for professionals, or lesson planning for educators) where AI tools – such as translators, ChatGPT, or image generators – can enhance outcomes. Start with small-scale experiments: use AI to summarize texts, visualize data, or generate initial content drafts, focusing on tasks where human oversight ensures responsible use (Zhou et al., 2025).

Through iterative refinement and critical evaluation, transform practice into competency. Continuously test outputs, adjust prompts, and troubleshoot errors (Chiu, 2025) while asking, Does this tool improve efficiency or accuracy? What safeguards prevent misuse? Measure results against goals (e.g., time saved, insights gained), and integrate feedback to optimize workflows. Crucially, anchor this process in ethical integration – maintaining transparency, auditing for bias, and preserving human judgment. Competency emerges when AI application becomes intentional, adaptive, and ethically embedded, turning literacy into measurable, context-aware excellence that drives superior outcomes. This maintains the core sequence: literacy → practice → critical refinement → ethical integration → competency, while emphasizing action and responsibility.

5. Future directions

AI literacy and AI companies are new to policymakers, researchers, educators, and students. In this editorial, we suggest five research directions in terms of questions

1. Mapping and optimizing AI literacy and AI competency development: What specific pedagogical interventions and practices most effectively bridge foundational AI literacy from the eight literacies to demonstrable competency? How does the development differ across learner profiles (e.g., age, profession, prior digital skills)?
2. The dynamics of interdisciplinary literacy integration for AI: How do the eight foundational literacies (mathematical, data, ethical, media, computational, linguistic, visual, and domain-specific) interact dynamically during the development of AI literacy and competency? Where are the critical synergy points or friction points, and how can curricula/training be designed to foster integrated understanding rather than siloed knowledge?
3. Ethical literacy as the active scaffold for competency: How does ethical literacy transition from theoretical awareness (“knowing the risks”) to an active, integrated practice (applying safeguards, auditing for bias, ensuring transparency) within the development and execution of AI competency? What frameworks and tools best support practitioners in making real-time ethical decisions while using AI tools under pressure?
4. Equity and access in the AI literacy-to-competency pipeline: How do disparities in foundational literacies (e.g., mathematical, data, linguistic, and computational access) create barriers to developing AI literacy and, subsequently, AI competency? What are the most effective strategies for mitigating these barriers and ensuring equitable pathways to AI competency across diverse populations (socioeconomic, cultural, linguistic)?
5. Culturally Responsive AI literacy & competency frameworks: How do cultural values, linguistic diversity, and local epistemologies shape the definition, acquisition, and application of AI literacy and competency? How can frameworks be adapted to ensure they are culturally sustaining rather than imposing techno-centric?

This editorial responds to one of the more popular globally education initiatives – AI literacy and competency – and encourages us to have a conversation about how to advance AI education. We look forward to learning more from your studies.

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